

1 Prove by induction that, for $n \geq 2$, $\sum_{r=2}^n \frac{(3-2r)3^r}{r(r-1)} = 9 - \frac{3^{n+1}}{n}$. [6]

2 Express $\frac{2}{r(r^2-1)}$ in partial fractions. [3]

Given that $S_N = \frac{1}{(4)(15)} + \frac{1}{(5)(24)} + \frac{1}{(6)(35)} + \dots + \frac{1}{(N)(N^2-1)}$, find S_N in terms of N .

Deduce the limit of S_N as $N \rightarrow \infty$. [4]

3 The functions f and g are defined by

$$f: x \mapsto 3 - x, \quad x < 3,$$

$$g: x \mapsto 2x^2, \quad x \in \mathbf{R}.$$

(i) Explain why the function gf^{-1} exists. [1]

(ii) Find $gf^{-1}(x)$. State the domain and the range of gf^{-1} . [4]

(iii) Describe a sequence of transformations that map the graph of $y = g(x)$ for $0 < x < 3$ onto the graph of $y = \frac{1}{2}gf^{-1}(-x)$ for $-3 < x < 0$. [2]

4 A gardener needs to spread 1500 kg of sand over his garden. He spreads 5 kg during the first day, and increases the amount he spreads each subsequent day by 2 kg.

(i) Find an expression for the mass of sand the gardener has spread by the end of the n th day. [2]

(ii) Deduce the minimum number of days required for him to spread the 1500 kg of sand. [3]

The gardener's neighbour also needs to spread 1500 kg of sand over his garden. He decides to spread 75 kg each day, but discovers that during each subsequent day, the amount of sand he can spread is 5% less than that of the previous day.

(iii) Find an expression for the mass of sand the neighbour has spread by the end of the n th day. [2]

(iv) Comment on the practicality of the approach taken by the neighbour. [1]

[Turn over]

- 5 In a certain country, the price of a brand new car of a particular make, manufactured on 1 January 1996, is \$32,000. According to a model of car pricing, the price P of the car (in \$) depreciates at a rate proportional to P when the car is t years old (as from 1 January 1996). Write down a differential equation relating P and t . [1]

By solving this differential equation, show that

$$P = 32000e^{-kt}$$

where k is a positive constant. [4]

A man purchased a used car of this particular make for \$2000, at the price predicted by the model, on 1 January 2006. Subsequently on 1 January 2007, the man sold the used car for \$800. Determine if the man sold his car below the price predicted by the model. [3]

- 6 With respect to the origin O , the position vectors of two points A and B are given by $\mathbf{a}$ and $\mathbf{b}$ respectively where $|\mathbf{a}| = 1$, $|\mathbf{b}| = 2$ and $\mathbf{a} \cdot \mathbf{b} = 1$. C is the point on AB such that $AC = 2CB$.

(i) Find the position vector of the point C in terms of $\mathbf{a}$ and $\mathbf{b}$. [1]

(ii) Show that the length of projection of OC onto OB is $\frac{3}{2}$ and deduce the position vector of the foot, N , of the perpendicular from C to the line OB . [4]

(iii) Find the value of λ such that C , D and N are collinear where $\overrightarrow{OD} = \mathbf{a} + \lambda\mathbf{b}$. [3]

- 7 The region R is the finite region bounded by the curve $y^2 + (x - 2)^4 = 1$.

(i) Sketch the curve and indicate clearly the region R . [2]

(ii) Find the area of R . [2]

(iii) The curve $y = (x - 1)^2$ divides R into two regions. Find the exact volume of the solid of revolution formed when the smaller region is rotated through 4 right angles about the x -axis. [5]

- 8 Two skew lines l_1 and l_2 are given by the equations

$$l_1 : \mathbf{r} = -8\mathbf{i} + 12\mathbf{j} + 16\mathbf{k} + s(\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}), s \in \mathbf{R} \text{ and}$$

$$l_2 : \mathbf{r} = 4\mathbf{i} + 6\mathbf{j} - 8\mathbf{k} + t(2\mathbf{i} - \mathbf{j} + 2\mathbf{k}), t \in \mathbf{R}.$$

The points P and Q lie on l_1 and l_2 respectively such that PQ is perpendicular to l_1 and l_2 .

- (i) Show that PQ is parallel to the vector $4\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}$. [2]
- (ii) Given that P is the point $(-8, 12, 16)$, find the position vector of Q . [3]
- (iii) Find the cartesian equation of the plane Π which contains l_1 and is parallel to l_2 . [2]
- (iv) Find a vector equation of l_3 , the reflection of l_2 in the plane Π . [3]

- 9 (a) Given that z is a non-zero solution of the equation $(8+i)z = (4-7i)z^*$, find the possible values of $\arg(z)$ correct to 3 decimal places. [5]

- (b) Find the exact roots of the equation $z^5 = -(16\sqrt{2})(1+i)$, expressing them in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [4]

On an Argand diagram the points which represent the above roots are rotated $\frac{1}{10}\pi$ radian anti-clockwise about the origin to obtain points A, B, C, D and E .

Find the equation whose roots are represented by A, B, C, D and E , giving your answer in the form $z^5 = -(16\sqrt{2})(p+iq)$ where p and q are real constants to be determined. [2]

- 10 (a) Find the following integrals.

(i) $\int \frac{x+1}{x^2+1} dx$, [3]

(ii) $\int (5x-3)\sqrt{x-3} dx$, using the substitution $u = \sqrt{x-3}$. [4]

- (b) Let $I = \int_0^{\frac{\pi}{2}} e^x \sin 2x dx$ and $J = \int_0^{\frac{\pi}{2}} e^x \cos 2x dx$.

Show using integration by parts, or otherwise, that

$$I = -2J \quad \text{and} \quad J = 2I - e^{\frac{\pi}{2}} - 1. \quad [3]$$

Hence find the exact values of I and J . [3]

[Turn over]

- 11 (a)** The curve C has the equation $2^{-y} = x$. The point A on C has x -coordinate a where $a > 0$.

Show that $\frac{dy}{dx} = -\frac{1}{a \ln 2}$ at A and find the equation of the tangent to C at A . [3]

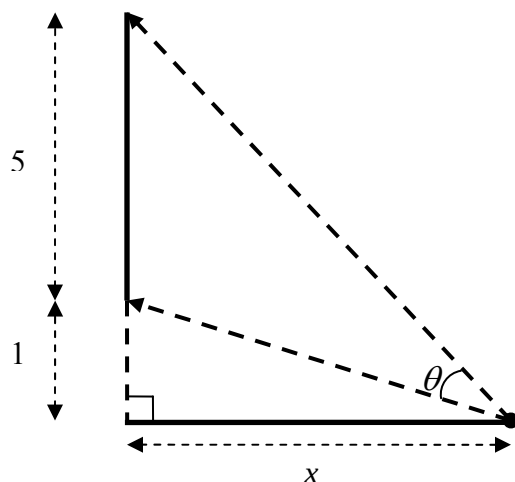
Hence find the equation of the tangent to C which passes through the origin. [2]

The straight line $y = mx$ intersects C at 2 distinct points.

Write down the range of values of m . [1]

- (b)** A movie theatre screen which is 5 m high, has its lower edge 1 m above an observer's eye.

The visual angle θ of the observer seated x m away is as shown in the diagram below.



- (i)** Show that $\theta = \tan^{-1}\left(\frac{6}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right)$. [1]

- (ii)** Find the exact distance the observer should sit to obtain the largest visual angle.

[You need not establish that the distance gives the largest visual angle.] [4]

- (iii)** Suppose that the observer is situated between 2 m and 15 m from the screen. Find, to the nearest degree, the smallest visual angle. [2]